Classify / Compare → DL

Siamese Network

Feature Matching and Detection

⇒ Intelligent Vision

feature-matching

feature could be any characteristic description of the image

"Matching"

↓

is sort to template

In any application

↓

You will be given a template

↓

And an image

↓

Aim of matching is to find location of template

within the image.

Applications :

1) Identifying position of objects
2) Facebook face tagging
3) Panoramic images
4) Autonomous vehicle

Simplest algorithm for Image matching

Take a window and Compare pixels of the image and the window.

This does not work in practice

Pixel intensities are not invariant to lighting condition & scaling, Rotation and other artifacts.

for image matching it is desired that features should have invariant properties.

Image → set of spatial pixels → set of spatial features

General Image matching Algorithm

Step1: Find a suitable feature set for the application.

Step 2: Represent given image as a feature set. Represent template as a feature set

Step 3: Compare the two feature images (or set) obtained in step 2.

Invariance properties that a "Good feature" should satisfy:

1) Localizability: The chosen feature should allow you to find the position of the template in the given image.

2) Invariance (Scale, Rotation or illumination)

↓

feature should not change if in the image desired object is scaled, rotated or intensity is increased.

scale ↓ Image taken from further distance.

3) Distinctiveness

↓

Two copies of template present in the image. Feature description for both the copies should be different (based on position, orientation in the image).

4) Robustness to Noise or Degradation:

↓

Image Noise or degradation should not effect features.

There is no such one feature which will do good in all four properties.

↓

We choose a feature as a trade off with respect to

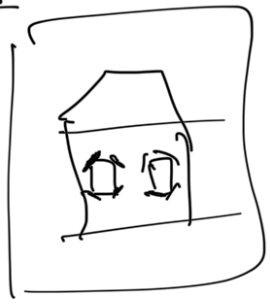
~~Utility~~ Computational requirements

Harris corner → DoG

Feature:

detecting windows
detecting object with concrete boundaries

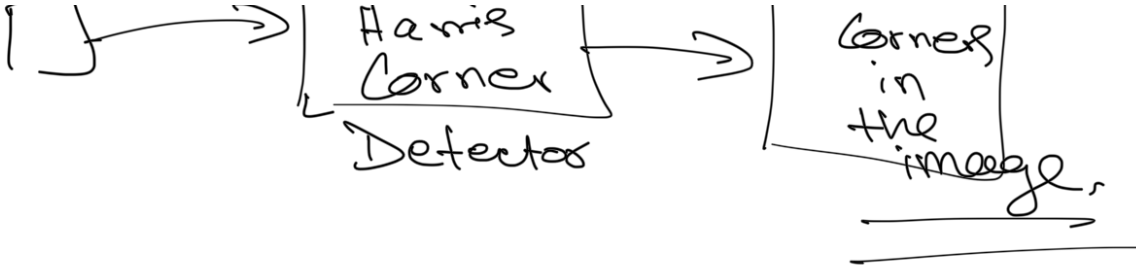
Corners in an image:



Corners feature Detection
- Localizability
- Invariance (illumination, Rotation, Scaling)

DoG

SIFT
- Distinctiveness
- descriptor
  - Robustness to noise (this would depend on application)


Harris
Corner
Detector
Corners
in
the
image.